

To prepare effectively for your upcoming BAE654C exam, it is crucial to understand the patterns established in the June/July 2025 examination paper and assignments. The examination typically requires answering **five full questions**, choosing one from each module. Most questions are at **Bloom's Level 2 (Understand/Explain)**, requiring you to illustrate concepts with "neat sketches" or sketches.

Here is a list of 10–15 important topics and questions per module that are likely to appear or might catch you off guard based on the syllabus and previous patterns.

Module 1: Basics

This module focuses on history, fundamental physics, and classification.

1. **Classification of Drones in India:** Be ready to classify them by weight (Nano, Micro, Small, Medium, Large) and explain their specific roles.
2. **HALE vs MALE vs TUAV:** Differentiate between these based on altitude, endurance, and strategic vs tactical roles.
3. **Generic UAV System Components:** Illustrate and explain the role of the GCS, payloads, data links, and launch/recovery subsystems.
4. **Newton's Laws in Flight:** Specifically, how the Third Law applies to propeller thrust and the First Law to hovering stability.
5. **Fundamental Forces Balance:** Explain the force equilibrium during **hovering** (Lift=Weight) vs **ascending** (Lift>Weight).
6. **6 Degrees of Freedom (DOF):** Define the three translational (surge, sway, heave) and three rotational (roll, pitch, yaw) movements.
7. **Stick Movements:** Explain how Mode 1 and Mode 2 transmitter configurations map to drone flight dynamics.
8. **The "DDD" Roles:** Explain why UAVs are preferred for "Dull, Dirty, and Dangerous" missions.
9. **History - The "Queen Bee":** Why is it called the "Mother of all Drones," and how did WWII influence UAV development?
10. **Swarm Technology:** Evaluate the "cost vs survivability" advantage of swarms in modern warfare.
11. **USV (Unmanned Surface Vehicles):** Illustrate its components and discuss its real-world advantages.
12. **Vortex Ring State:** Explain this dangerous state and why it occurs during rapid descent.
13. **Ground Effect:** Describe the increased lift experienced when flying close to the surface.

14. **Evolution of Control:** Discuss the transition from 20th-century radio control to modern autonomous intelligence.

Module 2: Components and Systems

This module deals with hardware, sensors, and software architecture.

1. **Microcontrollers vs Microprocessors:** Compare their specific roles and performance differences in drone operations.
2. **Arduino Sensors:** Explain their working principle and analyze real-world advantages/disadvantages.
3. **Flight Planning Process:** List the steps from mission objective definition to post-flight analysis.
4. **Pre-Flight Checklists:** Why are they mandatory for safe operations? List key items checked.
5. **Telemetry Systems:** Explain how real-time data (altitude, battery, GPS) is transmitted back to the GCS.
6. **Flight Controller (FC) Functions:** Describe how the FC handles stabilization, failsafes, and mode management.
7. **Fixed-Wing Drone Components:** Provide a neat labeled sketch of the essential airframe and control surfaces.
8. **Quadcopter Components:** Illustrate components like ESCs, BLDC motors, and the power distribution panel.
9. **Manual vs Autonomous Modes:** Differentiate between pilot-controlled flight and pre-programmed mission execution.
10. **Hybrid Flight Modes:** Explain how assisted control and failsafe mechanisms enhance safety.
11. **Navigation Sensors:** Detail the roles of the Magnetometer (compass) and Barometer (altitude).
12. **Obstacle Avoidance:** Discuss the types of sensors used to perceive the environment in real-time.

Module 3: Airworthiness

This module covers the legal and regulatory framework in India.

1. **Drone License Procedure:** Explain the step-by-step process of obtaining a Remote Pilot Certificate via the Digital Sky platform.

2. **NPNT (No Permission, No Take-off):** Describe how this software-firmware requirement ensures compliance before flight.
3. **DGCA Flying Zones:** Explain the operational permissions required for Green, Yellow, and Red zones.
4. **Certification Schemes:** Summarize the different types of certificates required for drone manufacturing and operation in India.
5. **Digital Sky Platform:** Discuss its impact on the growth of drone startups and regulated flight operations.
6. **NPNT Penalties:** Be aware of the fines for individuals and organizations for non-compliance.
7. **Drone Rules 2021:** Analyze how these rules transformed the Indian drone ecosystem compared to earlier regulations.
8. **Eligibility for Remote Pilots:** List the age, education, and training requirements for a pilot certificate.
9. **Pilot Examination Topics:** What subjects must a remote pilot master to pass the DGCA examination?.
10. **Small UAS Limitations:** Describe operating height and distance restrictions specifically for small categories.
11. **Airworthiness Definition:** Define what makes an aircraft “airworthy” in terms of design and safe condition.
12. **Drone Registration:** Explain the process of obtaining a Unique Identification Number.

Module 4: Structures and Propulsion Basics

Focus here is on physical design, power, and efficiency.

1. **Small Category Design Requirements:** Describe the specific design constraints for drones weighing 2kg to 25kg.
2. **Payload Configurations:** Discuss mounting, weight distribution, and common types (Camera, Sensor, Delivery).
3. **LiPo Battery Safety:** Explain the critical precautions for charging, storage, and handling to prevent fire hazards.
4. **Solar Powered UAV Principle:** Illustrate with a sketch how solar energy is used for propulsion and its current limitations.
5. **Hybrid Propulsion Advantages:** Explain why combining electric motors with IC engines increases fuel efficiency and range.
6. **Fixed-Wing vs Rotary-Wing Configurations:** Compare their efficiency, endurance, and ability to hover.

7. **Boundary Layer Concept:** Describe its effect on airflow over wings and its impact on UAV performance.
8. **Material Selection:** Why are carbon fiber and composites preferred over traditional metals for drone frames?.
9. **Battery Characteristics:** Discuss parameters like mAh, C-rating, and voltage in the context of flight endurance.
10. **Propeller Design:** How do pitch, diameter, and RPM affect thrust and energy consumption?.
11. **IC Engines vs Electric Motors:** Discuss when a designer would choose an internal combustion engine over a battery system.
12. **Frame Geometry:** Explain how the arm layout (X-frame, H-frame) impacts stability.

Module 5: Tuning, Testing, and Manufacturing

This module is about practical deployment and production.

1. **Manufacturing Constraints:** Detail the technical, material, and financial limitations faced during drone production.
2. **Ground Testing vs Flight Testing:** Differentiate between structural/calibration checks on the ground and performance checks in the air.
3. **Simulator Training Importance:** Why is virtual practice essential for advanced maneuvers and emergency prep?.
4. **Environmental Testing:** How do engineers evaluate performance in extreme wind, rain, and temperature?.
5. **Tuning and Calibration:** Explain the procedure for calibrating sensors like the IMU and GPS before a mission.
6. **Drone Repair Checklists:** Elaborate on the common points of failure and standard repair procedures.
7. **Software Testing:** Discuss the importance of testing Return-to-Home (RTH) and geofencing algorithms.
8. **Compliance Testing:** How is a drone verified against DGCA/FAA standards for transmission power and range?.
9. **Case Study Preparation:** Be ready to explain one real-time drone application (e.g., agriculture or search and rescue) in detail.
10. **PID Controllers:** Explain the role of these algorithms in processing sensor data to stabilize flight.
11. **Assembly Integration:** Discuss the challenges of integrating mechanical frames with electronic subsystems.

12. **Modularity and Upgradeability:** Why is modular design a key constraint and advantage in drone manufacturing?.